# ASHISH T VASANT

**FORWARD DEPLOYED / APPLIED AI ENGINEER**

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Portfolio: u4ar.github.io/ashish-ai-portfolio | GitHub: github.com/U4AR | github.com/voicekeyboarddev

## SUMMARY

Applied AI engineer with nearly 5.5 years at Bharat Electronics Limited, owning end-to-end prototypes across LLM systems, voice agents, RAG, computer vision, AR/VR, digital twins, drones, and sensor integration. Strong record of turning ambiguous operational requirements into working systems under constrained data, hardware, connectivity, privacy, and deployment conditions. Hands-on across Python, backend services, evaluation harnesses, local/open-source models, Google ADK, Gemini Live, GCP, Rust, Flutter, Unity, and edge/on-premise environments.

## CORE STRENGTHS

**AI & agents:** LLMs, multimodal models, RAG, retrieval, tool use, structured outputs, agentic workflows, Google ADK, context engineering, local/open and hosted models

**Systems & evals:** Inference optimization, benchmark/evaluation harnesses, latency/quality trade-offs, FastAPI, WebSockets, Docker, Linux, Git, failure analysis

**Cloud & deployment:** GCP, Cloud Run, Firebase, Firestore, local/on-premise and restricted environments, H100 inference, Android and Windows delivery

**Languages:** Python, Rust, JavaScript/TypeScript, Dart/Flutter, C#/Unity, OpenCV, PyTorch, TensorFlow, ROS, MATLAB

## PROFESSIONAL EXPERIENCE

**Deputy Engineer — Applied AI, Spatial Computing & Systems Prototyping** | Bharat Electronics Limited, Bengaluru | 2021–Present

- Owned cross-domain prototypes from ambiguous operational needs through architecture, implementation, sensor/model integration, stakeholder demonstrations, and iterative refinement in a defence/public-sector environment.
- Built an RFP item-finder and verification workflow that extracts products, clauses, and specifications, discovers candidates, and verifies evidence against requirements—an agentic enterprise search-and-check loop.
- Developed a local RAG pipeline for Indian-language agriculture documents, combining ingestion, domain retrieval, and regional-language answer generation for privacy-preserving knowledge access.
- Built smart-city perception across live CCTV/RTSP feeds: request-driven detection, Chandigarh ANPR, low-light vehicle detection, and person-to-3D mapping for a Unity room digital twin.
- Trained a YOLO hand-keypoint model from scratch for noisy monochrome spatial-display imagery; integrated it into natural interaction where standard RGB trackers were unsuitable.
- Created datacentre digital twins from SNMP data and Vision-LLM interpretation of dashboard imagery; translated imperfect operational sources into spatial monitoring views.
- Built AR/VR, HoloLens, ROS, ORB-SLAM, RealSense, LiDAR, AirSim, and drone systems under restricted connectivity/API access; contributed to two patent-applied AR workflows and mentored junior contributors.

## SELECTED AI SYSTEMS & OPEN-SOURCE PROJECTS

**GLM-5.2 (754B MoE) on 2×H100 — Inference Optimization & Evals** | github.com/U4AR/glm52-fast-2xh100 | 2026

- Engineered heterogeneous CPU/GPU serving, increasing measured single-stream decode from ~14.7 to ~40.5 tokens/sec (2.75×) through packed INT4 execution, top-2 expert substitution, MTP speculative decoding, and CUDA graphs.
- Fit CPU-side experts in ~250 GB RAM while retaining selected experts on GPU; delivered an OpenAI-compatible streaming API and browser UI.
- Built repeatable decode probes plus LiveBench and Terminal-Bench evaluation workflows to compare speed configurations against reasoning quality and regressions.

**LiveLingo — Real-Time Multilingual Voice Agent** | Flutter · FastAPI · Gemini Live · Google ADK · GCP | 2026

- Built a voice agent that autonomously converses and negotiates in a target language or coaches users with native script, translation, familiar-script transliteration, and pronunciation audio.
- Designed a low-latency PCM16/WebSocket pipeline with parallel translation/transliteration and three-way classification for echo rejection, user instructions, and counterpart speech; deployed with Cloud Run, Firebase, and Firestore.

**Voice Keyboard — Private On-Device Voice-to-Keystroke System** | Rust · Tauri · Gemma · llama.cpp | 2026

- Built an offline Windows app that converts speech into context-aware text and keyboard shortcuts in any focused application; no audio leaves the device.
- Integrated local multimodal inference, cursor/selection context, Win32 injection, GPU-aware setup, diagnostics, safety controls, and paired JSONL/WAV feedback capture.

**DoLess-Agents — Demonstration-to-Tool Browser Automation** | JavaScript · Chrome · Gemini · Ollama · Docker | 2026

- Built a Chrome extension that records/replays workflows and converts demonstrations into reusable AI-generated functions, reducing repeated visual-agent latency and cost.
- Integrated 12 tools across scraping, semantic search, scheduling, state, page modification, reporting, local/cloud reasoning, and computer use.

## EDUCATION & CREDENTIALS

**NIT Calicut — B.Tech, Mechanical Engineering** | 2016–2020 | CGPA 8.38/10

Robotics Interest Group member and mentor; ABU Robocon 2017; control systems, vision robotics, Series Elastic Actuators, MATLAB/Simscape.

**Certifications:** DeepLearning.AI Deep Learning Specialization · Columbia Computer Vision Basics · IBM Flutter & Dart Mobile Apps

**Languages:** English (fluent) · Malayalam (native) · Hindi (professional proficiency)